Small Lyapunov ratios and the structure of Berry–Esseen maximizers

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Abstract

We establish asymptotic and exact structural theorems for independent centered real arrays with small Lyapunov ratio. First, for any sequence of finite independent arrays whose Lyapunov ratios L_n tend to zero and whose Berry–Esseen ratios C_n tend to Esseen's constant $C_{\mathrm{E}} = (3+\sqrt{10})/(6\sqrt{2\pi})$, we prove that the summands are approximated coordinatewise by centered distributions supported on cosets of a common lattice $h_n\mathbb{Z}$ with summed squared and cubed Wasserstein errors $o(h_n^2)$ and $o(h_n^3)$, while the normalized zero-bias transport ratio $2B_n^2 W_1(S_n, S_n^*)/R_n$ converges to 1. Second, we prove that there exists an absolute constant $\varepsilon_0 > 0$ such that every global maximizer of the Berry–Esseen ratio over all N-summand independent arrays with $L < \varepsilon_0$ consists of identically distributed nondegenerate two-point summands and deterministic zero summands. To eliminate rounding residuals of arbitrarily small amplitude modulo $h_n\mathbb{Z}$, the proof combines variational contact equations on the topological supports with a relative conditional first-moment bound $\mathbb{E}(|T| \mid K = k) \geq c\mathbb{E}|T|$ valid for arbitrarily small $\mathbb{E}|T| > 0$, followed by a discrete Green-function concavity argument on the lattice.

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1 Introduction and main results

1.1 Asymptotic and exact structure at small Lyapunov ratio

Let $X_1, \dots, X_N$ be independent centered real random variables with positive total variance and finite third absolute moments, and set

$$S = \sum_{i=1}^{N} X_i, \qquad B^2 = \sum_{i=1}^{N} \mathbb{E}X_i^2 > 0, \qquad R = \sum_{i=1}^{N} \mathbb{E}|X_i|^3 < \infty, \qquad L = \frac{R}{B^3}, \qquad C = \frac{d_{\mathrm{K}}(S/B, Z)}{L}, \tag{1.1}$$

where $Z \sim N(0,1)$ and d_{K} denotes Kolmogorov distance. Deterministic zero summands are permitted. Define the Esseen constants

$$C_{\mathrm{E}} = \frac{3+\sqrt{10}}{6\sqrt{2\pi}}, \qquad p_* = \frac{4-\sqrt{10}}{2}, \qquad q_* = 1 - p_*, \qquad t_* = q_* - p_* = \sqrt{10} - 3,$$

$$r_* = p_*^2 + q_*^2, \qquad \beta_* = \frac{r_*}{\sqrt{p_* q_*}}, \qquad \nu_* = r_*^{-1},$$

and let W denote the centered unit-variance Esseen two-point random variable with $\mathbb{P}(W = \sqrt{q_*/p_*}) = p_*$ and $\mathbb{P}(W = -\sqrt{p_*/q_*}) = q_*$. For probability laws with finite jth moments, $W_j(X,Y) = (\inf \mathbb{E}|\widetilde{X} - \widetilde{Y}|^j)^{1/j}$ denotes the L^j-Wasserstein distance. The zero-biased distribution S^* is defined by $\mathbb{E}[Sf(S)] = B^2\mathbb{E}f'(S^*)$ [6], and satisfies the normalized transport inequality

$$\mathcal{B}(X) := \frac{2B^2 W_1(S, S^*)}{R} \le 1 \tag{1.2}$$

[5, 10], with equality if and only if all nonconstant summands share a single two-point distribution [3, Theorem 1.1].

Our first theorem characterizes arbitrary sequences of independent arrays that approach Esseen's constant C_{E} as $L_n \to 0$.

Theorem 1.1 (Asymptotic lattice and Esseen structure as $L_n \to 0$)**.** *For each $n \ge 1$, let $(X_{n,1}, \dots, X_{n,N_n})$ be a finite independent array satisfying (1.1), and suppose that*

$$L_n \longrightarrow 0, \qquad C_n \longrightarrow C_{\mathrm{E}}. \tag{1.3}$$

Then there exist lattice spacings $h_n \sim R_n/(r_* B_n^2)$ and centered distributions $X_{n,i}^{\mathrm{lat}}$ supported on cosets of $h_n\mathbb{Z}$ such that

$$\sum_{i=1}^{N_n} W_2^2(X_{n,i}, X_{n,i}^{\mathrm{lat}}) = o(h_n^2), \qquad \sum_{i=1}^{N_n} W_3^3(X_{n,i}, X_{n,i}^{\mathrm{lat}}) = o(h_n^3), \tag{1.4}$$

and the zero-bias transport ratio satisfies $\mathcal{B}(X_n) \to 1$.

Furthermore, there exist subsets $I_n \subseteq \{1, \dots, N_n\}$ of cardinality $m_n = |I_n|$ satisfying $m_n L_n^2/\beta_*^2 \to 1$ and carrying $1 - o(1)$ of both B_n^2 and R_n, together with a sign choice $\pm W$ for each row, such that the padded i.i.d. Esseen distributions

$$Q_{n,i} = \mathrm{Law}\left(\frac{B_n(\pm W)}{\sqrt{m_n}}\right) \quad (i \in I_n), \qquad Q_{n,i} = \delta_0 \quad (i \notin I_n)$$

satisfy

$$\sum_{i=1}^{N_n} W_2^2(X_{n,i}, Q_{n,i}) = o(B_n^2), \qquad \sum_{i=1}^{N_n} W_3^3(X_{n,i}, Q_{n,i}) = o(R_n).$$

Notice the distinction between the two approximations in Theorem 1.1: the common-lattice comparators $X_{n,i}^{\mathrm{lat}}$ in (1.4) achieve error smaller than the microscopic lattice spacing h_n, but may be supported on more than two points of their respective cosets of $h_n\mathbb{Z}$; the two-point comparators $Q_{n,i}$ achieve error smaller than the macroscopic array scales B_n and $R_n^{1/3}$ because a rare multi-step lattice excursion can contribute $o(R_n)$ to the third moment while having size of order h_n.

When each array in the sequence is an actual global maximizer of the Berry–Esseen ratio at its own number of summands N, the common-lattice support and padded i.i.d. two-point structure hold without approximation error for every sufficiently late array in the sequence (once $L_n < \varepsilon_0$). For each $N \geq 1$, let C_N denote the supremum of C in (1.1) over all independent centered arrays of at most N summands.

Theorem 1.2 (Exact structure of small-Lyapunov global maximizers)**.** *There exists an absolute constant $\varepsilon_0 > 0$ such that every independent array $(X_1, \ldots, X_N)$ attaining C_N with $L < \varepsilon_0$ consists of identically distributed nondegenerate two-point summands and deterministic zero summands. Equivalently, along any sequence of global maximizers of C_{N_n} with $L_n \to 0$, every sufficiently late array is a padded i.i.d. two-point array.*

1.2 Open questions and relationship to companion papers

Let us clarify how Theorems 1.1 and 1.2 relate to the companion papers in this collection and to the open finite-N conjectures:

(1) *Qualitative exact structure versus explicit value cutoff:* The companion paper Bell [2] proves the explicit value bound $\Delta \leq C_{\mathrm{E}} L$ for every independent array with $L \leq \exp(-\exp(97))$ by selecting a penalized hypothetical counterexample above C_{E}. By contrast, at any finite N, the maximum C_N is strictly smaller than C_{E} on binomial arrays [8]. Theorem 1.2 classifies the actual maximizers attaining C_N when $L < \varepsilon_0$; its threshold $\varepsilon_0 > 0$ is established by compactness and is not effective. Instead of depending on Bell [2], our proof of Theorem 1.1 uses Shevtsova's published asymptotic upper bound

$$d_{\mathrm{K}}(S/B, Z) \leq C_{\mathrm{E}} L + 0.2538 L^{4/3} \qquad (L \leq 10^{-3}) \tag{1.5}$$

[9, Corollary 4.18, p. 303] and extracts the nonnegative moment and tail losses directly from her smoothing proof.

(2) *Fixed-N conjectures for unrestricted L:* Because adjoining zero summands increases N without changing L, an N-summand global maximizer could a priori use a bounded number of active summands with $L \geq \varepsilon_0$ even when N is large. Proving that $L_n \to 0$ automatically along $N_n \to \infty$—or proving that every global maximizer of C_N is a padded i.i.d. two-point array for *all* $N \geq 1$—remains open and would imply $C_{\mathrm{ind}} = C_{\mathrm{E}}$.

1.3 Overview of the proofs

The proofs of Theorems 1.1 and 1.2 proceed through seven sections:

- **Existence of fixed-N maximizers (Section 2):** Lemma 2.1 proves that C_N is attained for every $N \geq 1$ and that every maximizing sequence normalized to $B = 1$ has a subsequence converging coordinatewise in W_3.

- **Retained smoothing losses and the Esseen core (Section 3):** Inspecting the Prawitz smoothing proof of Shevtsova [9, pp. 289–303] under $L_n \to 0$ and $C_n \to C_{\mathrm{E}}$ shows that the individual moment deficits $\delta_i = \sqrt{10}\rho_i - |\kappa_i| - 3b_i v_i$, the truncation tails above $R^{1/6}$, and the orientation cancellation $\sum_i |\kappa_i| - |\sum_i \kappa_i|$ are all $o(R)$, while $\sum_i \sigma_i^3/R \to 1/\beta_*$ (Lemma 3.1). A high-frequency Prawitz window lower bound (Lemma 3.2) then forces the variance distribution of σ_i/R to concentrate at $1/\beta_*$, yielding a core I_n of $m_n \sim \beta_*^2/R_n^2$ summands uniformly close in W_3 to scaled Esseen two-point variables (Proposition 3.3).

- **First nonzero Fourier peak and integer lifts (Sections 4–5):** In gap units $h_n \sim \nu_* R_n$, locating the maximum of the core characteristic function near 2π allows us to decompose each rescaled summand as $Y_i = J_i + Z_i$, where $J_i = K_i - \mathbb{E}K_i$ ($K_i \in \mathbb{Z}$) is a centered integer variable and Z_i is a centered rounding residual with $Q = \sum_i \mathbb{E}Z_i^2 = O(1)$ and $\mathrm{Cov}(K, T) = o(\sqrt{M})$ (Lemma 4.1). A joint local limit theorem for $((K-\mu)/\sqrt{A}, T)$ (Lemma 5.2) and the closed-threshold envelope (Lemma 5.3) then force $\mathbb{E}\,\mathrm{dist}(S - \vartheta_n, \mathbb{Z}) \to 0$, which implies both $\mathcal{B}(X_n) \to 1$ (by Kantorovich–Rubinstein duality against a sawtooth test function) and $Q = \sum_i \mathbb{E}Z_i^2 \to 0$, completing Theorem 1.1.

- **Elimination of rounding residuals for global maximizers (Sections 6–8):** For a global maximizer of C_N, the coordinatewise variational contact inequality (6.1) (adapting He and Cheng [7, Section 2.1]) bounds all supports by $|Y_i| \le H = O(V^{1/3})$ and forces the uniform residual bound $\eta_n = \max_i \|Z_i\|_\infty \to 0$ (Lemma 6.2), which implies $Q = o(t)$ whenever $t = \mathbb{E}|T| > 0$. Rescaling slightly to place the full characteristic product at an exact local maximum at frequency 2π gives $|\mathrm{Cov}(K, T)| \le KHQ$ (Lemma 7.1). Using a Khintchine truncation multiplier bound (Lemma 7.2) and a two-copy conditional Fourier expansion, Lemma 7.3 proves the *relative* conditional first-moment bounds $ct \le \mathbb{E}(|T| \mid K = k) \le Kt$ and $|\mathbb{E}(T \mid K = k)| = o(t)$, valid even if $t = \mathbb{E}|T|$ tends to zero arbitrarily fast. Replacing $Y_i = J_i + Z_i$ by the pure lattice summands J_i at the same N then increases the Berry–Esseen ratio by at least $c't$ whenever $t > 0$, forcing $t = 0$ and hence $Z_i = 0$ almost surely for every i (Proposition 8.2).

- **Discrete Green-function concavity and reduction to two points (Section 9):** Once all summands lie on cosets of $h_n\mathbb{Z}$, a local Edgeworth expansion of the deleted point mass differences shows that the second difference $\Delta^2 g_i(u)$ of the contact function on the lattice coset is strictly negative between support extremes. Discrete Green-function inversion then rules out any support span ≥ 2, reducing every nonzero summand to two adjacent lattice points (Lemma 9.1), where the common-diameter classification of Bell [1, Theorem 1.1] completes the proof of Theorem 1.2.

2 Attainment of the finite-N supremum

Writing $F_S(y) = \mathbb{P}(S \le y)$ and $F_S^-(y) = \mathbb{P}(S < y)$, continuity of Φ gives

$$d_{\mathrm{K}}(S/B, Z) = \sup_{y \in \mathbb{R}} \max\{F_S(y) - \Phi(y/B),\, \Phi(y/B) - F_S^-(y)\}, \tag{2.1}$$

and reflection $X_i \mapsto -X_i$ interchanges the two one-sided branches.

Lemma 2.1 (Existence and W_3 compactness of fixed-N maximizers)**.** *For each integer $N \ge 1$, the supremum C_N is attained (with zero summands allowed). Every sequence of N-summand arrays with $B = 1$ and ratio tending to C_N has a subsequence converging coordinatewise in W_3 to a maximizing array.*

Proof. A single symmetric two-point summand padded with $N-1$ zeros achieves $c_0 = \Phi(1) - 1/2 > 0$, so any array with ratio $\geq c_0$ and $B = 1$ satisfies $N^{-1/2} \leq R \leq 1/c_0$. In the product weak topology, centered N-tuples with $B = 1$ and $R \leq 1/c_0$ form a compact set (by tightness and uniform integrability of first and second moments), on which $d_{\mathrm{K}}(S, Z)$ is continuous and R is lower semicontinuous. Along an extremizing subsequence, maximality and lower semicontinuity of each $\mathbb{E}|X_i|^3$ force $\lim_k \mathbb{E}|X_{k,i}|^3 = \mathbb{E}|X_i|^3$ for every $1 \leq i \leq N$; weak convergence together with convergence of third absolute moments is equivalent to convergence in W_3. □

3 Retained smoothing losses and the Esseen core

In this section we establish the second assertion of Theorem 1.1: any sequence of arrays satisfying (1.3) concentrates in W_3 around a padded i.i.d. Esseen array. Normalize total variance to $B = 1$ and omit the row index n, so that $R = L \to 0$. For each summand X_i, put $v_i = \mathbb{E}X_i^2$, $\sigma_i = \sqrt{v_i}$, $\rho_i = \mathbb{E}|X_i|^3$, $\kappa_i = \mathbb{E}X_i^3$, $b_i = \mathbb{E}|X_i|$, and define the scalar moment deficit

$$\delta_i = \sqrt{10}\rho_i - |\kappa_i| - 3b_i v_i.$$

Lemma 3.1 (Retained moment, tail, and orientation losses)**.** *Under (1.3), as $R \to 0$,*

$$\sum_{i=1}^N \delta_i = o(R), \qquad \sum_{i=1}^N \mathbb{E}\big[|X_i|^3 \mathbf{1}_{\{|X_i| > R^{1/6}\}}\big] = o(R), \qquad \sum_{i=1}^N |\kappa_i| - \left|\sum_{i=1}^N \kappa_i\right| = o(R), \tag{3.1}$$

and $\sum_{i=1}^N \sigma_i^3/R \to 1/\beta_*$. Moreover, after reflecting the array if necessary so that $\sum_{i=1}^N \kappa_i \geq 0$, let $P_i^* = \mathrm{Law}(\sigma_i W)$ be the positively oriented Esseen two-point distribution of variance v_i. Then

$$\sum_{i=1}^N W_3^3(X_i, P_i^*) = o(R). \tag{3.2}$$

Proof. We first establish quantitative stability in Shevtsova's scalar moment inequality $\delta(X) \geq 0$ [9, Theorem 3.1, pp. 256–262]. Put $s = \sqrt{p_* q_*}$, $a = \sqrt{q_*/p_*}$, $b = \sqrt{p_*/q_*}$, and $t = t_*$. For a centered unit-variance variable X oriented to have $\mathbb{E}X^3 \geq 0$, set $Q(x) = -6sx^2 + 6tx + 6s$ and $f(x) = x^3 - \sqrt{10}|x|^3 + 3|x|$. Direct factorization gives

$$Q(x) - f(x) = \begin{cases} (\sqrt{10} - 1)(x - a)^2(x + 2s), & x \geq 0, \\ (\sqrt{10} + 1)(-x - b)^2(-x + 2s), & x \leq 0. \end{cases}$$

Since $\mathbb{E}Q(X) = 0$ and $\mathbb{E}f(X) = -\delta(X)$, we have $\delta(X) = \mathbb{E}[Q(X) - f(X)] \geq c_0 \mathbb{E}[\mathrm{dist}(X, \{-b, a\})^3]$ with $c_0 = 2p_*(\sqrt{10} - 1) > 0$. Rounding X to a on $[0, \infty)$ and to $-b$ on $(-\infty, 0)$ at cubic cost $e \leq \delta(X)/c_0$ and then matching the atom probabilities to p_*, q_* gives $W_3^3(X, W) \leq 4e + 4(a+b)^2 e^{1/3}$. Rescaling each summand by σ_i and applying Hölder's inequality yields, for individually oriented Esseen targets P_i^{oriented},

$$\frac{\sum_{i=1}^N W_3^3(X_i, P_i^{\mathrm{oriented}})}{R} \leq \frac{4}{c_0}\bar{\delta} + 4(a+b)^2 c_0^{-1/3} \bar{\delta}^{1/3}, \qquad \bar{\delta} = \frac{1}{R}\sum_{i=1}^N \delta_i. \tag{3.3}$$

Next, we retain the nonnegative moment and truncation losses in the smoothing proof of Shevtsova [9, pp. 289–296]. With truncation level $U = R^{1/6}$, set $Y_i = X_i \mathbf{1}_{\{|X_i| \leq U\}}$, $Z_i = Y_i - \mathbb{E}Y_i$, and $T_i = \mathbb{E}[|X_i|^3 \mathbf{1}_{\{|X_i| > U\}}]$. By Shevtsova [9, Lemma 4.3 and I_{14}–I_{18}, pp. 272–295], $\Delta \leq \phi(0)(I_{16} +$

$R/2) + O(R^{7/6})$, where $I_{16} = \frac{\sqrt{10}}{6}R - \frac{1}{6}\sum_i \delta(Z_i) + I_{17} - I_{18}$, $I_{17} = \frac{\sqrt{10}}{6}\sum_i(\mathbb{E}|Z_i|^3 - \mathbb{E}|Y_i|^3)$, and $I_{18} = \frac{\sqrt{10}-1}{6}\sum_i T_i$. Since $|\mathbb{E}Y_i| \le T_i/U^2$ gives $\mathbb{E}|Z_i|^3 - \mathbb{E}|Y_i|^3 \le \frac{1}{3}T_i$ for $R \le 1/4096$, we obtain

$$\Delta \le C_{\mathrm{E}}R - \frac{\phi(0)}{6}\sum_{i=1}^{N}\delta(Z_i) - \frac{\phi(0)(\sqrt{10}-1)}{12}\sum_{i=1}^{N}T_i + O(R^{7/6}).$$

Under (1.3), both sums must be $o(R)$, which proves the first two relations in (3.1).

To prove the third relation in (3.1), write $K_3 = \sum_i \kappa_i$, $\nu = 1 + \sum_i b_i v_i/R \in [1,2]$, and $T = 2\pi/(\nu R)$. On every compact frequency interval, $\chi(u) = \mathbb{E}e^{\mathrm{i}uS} = e^{-u^2/2}(1 - \mathrm{i}K_3u^3/6) + o(R)$, so dominated convergence in the low-frequency Prawitz integral of Shevtsova [9, Lemma 4.1 and (4.4)–(4.7), pp. 263–267] replaces $\sum_i|\kappa_i|$ by $|K_3|$:

$$\Delta \le \phi(0)\left\{\frac{R}{2} + \frac{1}{2}\sum_{i=1}^{N}b_iv_i + \frac{|K_3|}{6}\right\} + o(R) = C_{\mathrm{E}}R - \frac{\phi(0)}{6}\left\{\sum_{i=1}^{N}\delta_i + \sum_{i=1}^{N}|\kappa_i| - |K_3|\right\} + o(R), \quad (3.4)$$

forcing $\sum_i|\kappa_i| - |K_3| = o(R)$ and proving (3.2) via (3.3). Finally, Theorem 4.14 of Shevtsova [9, pp. 300–303] gives $\Delta \le cR + K(c)\sum_i\sigma_i^3 + O_c(R^{4/3})$ for every fixed $c > 2\phi(0)/3$, where $K(C_{\mathrm{E}}) = 0$ and $K'(C_{\mathrm{E}}) = -\beta_*$; varying c on both sides of C_{E} forces $\sum_i\sigma_i^3/R \to 1/\beta_*$. □

Lemma 3.2 (High-frequency Fourier window and subset variance control)**.** *For every fixed $\epsilon > 0$ and sufficiently small fixed $t_0 > 0$, set*

$$T' = (1+\epsilon)\frac{2\pi}{\nu R}, \qquad H = 2\int_{t_0}^{1}|K(u)|\,|\chi(T'u)|\,du,$$

where $K(u)$ is Prawitz's smoothing kernel. Under (1.3), $\nu \to \nu_*$ and

$$\liminf_{n\to\infty}\frac{H}{R} \ge \frac{\phi(0)\nu_*\epsilon}{2(1+\epsilon)} > 0. \quad (3.5)$$

Consequently, there exist $\eta > 0$ and $e_n \to 0$ such that for every subset of indices $A \subseteq \{1,\dots,N_n\}$,

$$\sum_{i\in A}v_i \le \eta^{-1}\frac{\sum_{i\in A}\rho_i}{R} + e_n, \quad (3.6)$$

and $\sum_{i=1}^{N_n}W_2^2(X_i,P_i^*) \to 0$.

Proof. Applying Prawitz's smoothing inequality at the enlarged cutoff T' bounds the low-frequency integral by $\phi(0)|K_3|/6 + o(R)$ and the Gaussian cutoff term by $\phi(0)\nu R/[2(1+\epsilon)] + O(R^2)$; comparing with $\Delta \sim C_{\mathrm{E}}R$ yields (3.5). Since $|\chi_A(u)| \le \exp(-V_Au^2/2 + 2R_A|u|^3/3)$ for $V_A = \sum_{i\in A}v_i$ and $R_A = \sum_{i\in A}\rho_i$, on $u \in [a/R, C/R]$ with $\eta \le 3/(8C)$ any subset with $R_A/V_A \le \eta R$ has $|\chi(u)| \le \exp(-a^2V_A/(4R^2))$, which combined with (3.5) gives (3.6). Applying (3.6) to the exceptional set $\{i : W_3^3(X_i,P_i^*) > \zeta\rho_i\}$ and Hölder's inequality on its complement proves $\sum_i W_2^2(X_i,P_i^*) \to 0$. □

Proposition 3.3 (Concentration at a single variance scale)**.** *The i.i.d. Esseen approximation of Theorem 1.1 holds. Moreover, in gap units $h = \nu_*R$, there exists a subset $I \subseteq \{1,\dots,N_n\}$ of cardinality $M = |I|$ carrying $1 - o(1)$ of both total variance and R such that*

$$MR^2 \longrightarrow \beta_*^2, \qquad \sup_{i\in I}W_3(\mathrm{Law}(X_i/h), \mathrm{Law}(B_{p_*} - p_*)) \longrightarrow 0, \quad (3.7)$$

and in these units the rescaled total variance V and third-moment sums P, κ satisfy $V \sim p_*q_*M$, $P/V \to r_*$, and $\kappa/V \to t_*$.

Proof. Consider the variance-weighted probability measures $\mu_n = \sum_i v_i \delta_{\sigma_i/R}$, whose means $\sum_i \sigma_i^3/R$ converge to $s_0 = 1/\beta_*$, and which have no limiting mass at 0 by (3.6). For each $\epsilon > 0$, (3.5) supplies a frequency $u_n = w_n/R$ with $\limsup w_n \le 2\pi(1+\epsilon)/\nu_*$ and $|\chi(u_n)| \ge c_\epsilon R$, so $\sum_i (1 - |\chi_i(u_n)|^2) = O_\epsilon(\log(1/R))$. Transferring to the Esseen targets P_i^* via $\sum_i W_2^2(X_i, P_i^*) \to 0$ shows that any weak limit μ of μ_n satisfies $\int \sin^2((a+b)zw_\epsilon/2)\, d\mu(z) = 0$, so $\operatorname{supp}\mu \subset [s_0/(1+\epsilon), \infty)$. Letting $\epsilon \downarrow 0$ gives $\operatorname{supp}\mu \subset [s_0, \infty)$, which together with $\int z\, d\mu \le s_0$ forces $\mu = \delta_{s_0}$ and $\int |z - s_0|\, d\mu_n \to 0$. Restricting to a slowly shrinking neighborhood of s_0 and discarding indices with large $W_3(\mathrm{Law}(X_i/h), \mathrm{Law}(B_{p_*} - p_*))$ yields (3.7). □

4 First nonzero Fourier peak and integer lifts

Lemma 4.1. *At a spacing h_n asymptotic to $\nu_* R_n/B_n^2 = \nu_* L_n B_n$ (so $h_n \sim \nu_* R_n$ under the normalization $B_n = 1$), the rescaled summands $Y_i = X_i/h_n$ decompose as*

$$Y_i = J_i + Z_i, \qquad J_i = K_i - \mathbb{E}K_i, \qquad K_i \in \mathbb{Z}, \qquad \mathbb{E}Z_i = 0, \qquad |Z_i| \le 1,$$

where the pairs (K_i, Z_i) are independent across i, $\sup_{i\in I} W_3(K_i, \mathrm{Bern}(p_*)) \to 0$ (up to integer shifts) on the core I of size M from Proposition 3.3, $\sum_{i\notin I}(\mathrm{Var}\, K_i + \mathbb{E}|J_i|^3) = o(M)$, and for $K = \sum_i K_i$ and $T = \sum_i Z_i$,

$$Q = \mathrm{Var}\, T = O(1), \qquad \mathrm{Cov}(K, T) = o(\sqrt{M}), \qquad A = \mathrm{Var}\, K \sim p_* q_* M. \tag{4.1}$$

Proof. Taking $0 < \epsilon < 1/2$ in Lemma 3.2, the frequency window in gap units lies inside $(0, 4\pi)$, where the core product $\prod_{i\in I} |\mathbb{E}e^{\mathrm{i}sY_i}|$ is exponentially small outside a neighborhood of 2π and attains its local maximum at $s_0 \to 2\pi$ with $\prod_{i=1}^N |\mathbb{E}e^{\mathrm{i}s_0 Y_i}| \ge c > 0$. Rescaling h_n by $2\pi/s_0 \to 1$ places this core maximum at 2π.

Aligning each summand by $\theta_i = (2\pi)^{-1} \arg \mathbb{E}e^{2\pi \mathrm{i} Y_i}$, write $Y_i = \theta_i + K_i + E_i$ with $K_i \in \mathbb{Z}$ and $E_i \in [-1/2, 1/2]$. Then $\mathbb{E}\sin(2\pi E_i) = 0$, $r_i = \mathbb{E}\cos(2\pi E_i) > 0$, and $\sum_i \mathbb{E}E_i^2 \le \frac{1}{8}\sum_i (1 - r_i) = O(1)$. Setting $Z_i = E_i - \mathbb{E}E_i$ gives $Q = \sum_i \mathbb{E}Z_i^2 = O(1)$. Stationarity of the core log-modulus at 2π is $\sum_{i\in I} \mathbb{E}[Y_i \sin(2\pi E_i)]/r_i = 0$; since $r_i \to 1$ uniformly on I and $|\sin(2\pi E_i) - 2\pi E_i| \le K|E_i|^3$, this implies $\mathrm{Cov}(K, T) = \mathrm{Cov}(\sum_i Y_i, T) - Q = o(\sqrt{M})$, proving (4.1). □

5 Joint integer and residual limit theorems

Under the decomposition $Y_i = J_i + Z_i$ of Lemma 4.1, write $J_i = K_i - \mathbb{E}K_i$, $K = \sum_{i=1}^N K_i$, $\mu = \mathbb{E}K$, $T = \sum_{i=1}^N Z_i$, $A = \mathrm{Var}\, K$, $Q = \mathrm{Var}\, T = O(1)$, $S = K - \mu + T$, and $V = \mathrm{Var}\, S$. Because the core I of size M converges uniformly in W_3 to $\mathrm{Bern}(p)$ ($p = p_*$) and carries all but $o(M)$ of the integer variance and third absolute moments, while $\mathrm{Cov}(K, T) = o(\sqrt{M})$, we have

$$A \sim p(1-p)M, \qquad V = A + o(\sqrt{A}), \qquad \gamma = \frac{\sum_{i=1}^N \mathbb{E}J_i^3}{A} \longrightarrow 1 - 2p. \tag{5.1}$$

Lemma 5.1 (Global Edgeworth and local estimates for the integer sum)**.** *Uniformly over all $j \in \mathbb{Z}$, with $x_j = (j - \mu)/\sqrt{A}$,*

$$\mathbb{P}(K \le j) = \Phi(x_j) + \frac{\phi(x_j)}{\sqrt{A}} \left\{ \frac{1}{2} + \frac{\gamma}{6}(1 - x_j^2) \right\} + o(A^{-1/2}), \tag{5.2}$$

$$\sqrt{A}\, \mathbb{P}(K = j) = \phi(x_j) + o(1), \tag{5.3}$$

and uniformly over all $k \in \mathbb{Z}$,

$$\mathbb{E}[T^2 \mathbf{1}_{\{K=k\}}] \le \frac{K_0 Q}{\sqrt{A}}. \tag{5.4}$$

All three estimates remain uniform after deleting any fixed number of summands.

Proof. On the core I, every K_i has atom probabilities at 0 and 1 bounded below, so after deleting a bounded number of summands the remaining product satisfies the torus damping bound

$$|\chi_{\text{deleted}}(u)| \le K \exp(-cM \sin^2(u/2)) \qquad (|u| \le \pi). \tag{5.5}$$

Expanding $\log \chi_I(u) = -Wu^2/2 - \mathrm{i}\kappa_I u^3/6 + O(M(\epsilon_n|u|^3 + |u|^4))$ near $u = 0$, inverting against the Gaussian–Hermite lattice sequence via Euler–Maclaurin summation, and convolving with the outside block (which has variance and third-moment sum $o(A)$) yields (5.2) and (5.3). For (5.4), expanding the Fourier transform of $T^2 \mathbf{1}_{\{K=k\}}$ into diagonal terms (bounded in sum by Q) and off-diagonal terms (with factors $|\mathbb{E}[Z_i e^{\mathrm{i}uJ_i}]| \le |u|\sqrt{\mathrm{Var}(K_i)\mathbb{E}Z_i^2}$) and integrating $KQ(1 + Au^2)e^{-cAu^2}$ over $[-\pi, \pi]$ gives the uniform bound $K_0 Q/\sqrt{A}$. □

Lemma 5.2 (Asymptotic independence and joint local limit)**.** *Along any subsequence for which $T \Rightarrow U$, we have $((K - \mu)/\sqrt{A}, T) \Rightarrow (G, U)$, where $G \sim N(0, 1)$ is independent of U and $\mathbb{E}U = 0$. Moreover, for every integer sequence k with $(k - \mu)/\sqrt{A} \to x \in \mathbb{R}$,*

$$\sqrt{A}\,\mathbb{P}(K = k,\ T \in du) \Longrightarrow \phi(x)\,\mathrm{Law}(U)(du). \tag{5.6}$$

Proof. For fixed $u, v \in \mathbb{R}$ and $s = u/\sqrt{A}$, expanding $\sum_{i \in I} \log(\mathbb{E}e^{\mathrm{i}sJ_i + \mathrm{i}vZ_i}/\mathbb{E}e^{\mathrm{i}vZ_i})$ through second order in s and using $\mathrm{Cov}(K, T) = o(\sqrt{A})$ gives $\mathbb{E}e^{\mathrm{i}s(K-\mu)+\mathrm{i}vT} = e^{-u^2/2+o(1)}\mathbb{E}e^{\mathrm{i}vT}$, proving joint weak convergence. To prove (5.6), invert the Fourier transform in the first coordinate at k. Since $\sup_{i \in I} \mathbb{E}Z_i^2 \to 0$, on a fixed small cell $|s| \le c_0$ the twisted core factors $\mathbb{E}e^{\mathrm{i}sJ_i + \mathrm{i}vZ_i}$ converge uniformly in C^2 to the centered Bern(p) factor, are bounded away from zero, and have log-modulus second derivatives at most $-c$ while their log-modulus first derivatives at $s = 0$ sum to at most $K_v\sqrt{A}$ by Cauchy–Schwarz. Thus

$$\left|\prod_{i \in I} \mathbb{E}e^{\mathrm{i}sJ_i + \mathrm{i}vZ_i}\right| \le \exp(K_v|s|\sqrt{A} - cAs^2) \qquad (|s| \le c_0),$$

whereas on $c_0 \le |s| \le \pi$ the uniformly binary-like twisted core product is exponentially small in M. Substituting $s = w/\sqrt{A}$, dominated convergence and tightness from (5.4) together with (5.3) yield (5.6). □

Lemma 5.3 (Asymptotic threshold envelopes)**.** *If $y/\sqrt{A} \to x \in \mathbb{R}$, let $j = \lfloor y + \mu \rfloor$ and suppose along a subsequence that $\theta = y + \mu - j \to \theta_0 \in [0, 1]$ and $T \Rightarrow U$. Then*

$$\limsup_{n \to \infty} \sqrt{A}\,\{F_S(y) - \Phi(y/\sqrt{V})\} \le \phi(x)\left\{\frac{1}{2} + \frac{1 - 2p}{6}(1 - x^2) - \mathbb{E}\{\theta_0 - U\}\right\}, \tag{5.7}$$

$$\liminf_{n \to \infty} \sqrt{A}\,\{F_S^-(y) - \Phi(y/\sqrt{V})\} \ge \phi(x)\left\{-\frac{1}{2} + \frac{1 - 2p}{6}(1 - x^2)\right\}, \tag{5.8}$$

where $\{w\} = w - \lfloor w \rfloor$ denotes fractional part. If $|y|/\sqrt{A} \to \infty$, then $\sqrt{A}\,|F_S^{\pm}(y) - \Phi(y/\sqrt{V})| \to 0$.

Proof. Decomposing $F_S(y) - F_K(j) = \sum_{\ell \in \mathbb{Z}}\{\mathbb{P}(K = j + \ell, T \le \theta - \ell) - \mathbf{1}_{\{\ell \le 0\}}\mathbb{P}(K = j + \ell)\}$, truncating at $|\ell| \le H$ via (5.4) (which bounds the tail by $KQ/(H\sqrt{A})$), applying (5.6) and Portmanteau on closed half-lines, and sending $H \to \infty$ gives $\phi(x)\mathbb{E}\lfloor \theta_0 - U \rfloor = \phi(x)(\theta_0 - \mathbb{E}\{\theta_0 - U\})$ (since $\mathbb{E}U = 0$). Combining this with (5.2) at $j = y + \mu - \theta$ proves (5.7), and open half-lines similarly prove (5.8). □

5.1 Modular concentration and proof of Theorem 1.1

For a sequence satisfying (1.3) with $p = p_*$, we have $P/V \to r_*$ and $\kappa/V \to t_*$ by Proposition 3.3. The negative strict branch (5.8) has normalized upper limit $\phi(0)(1/2 - t_*/6)/r_* < C_{\mathrm{E}}$, while the positive closed branch (5.7) has unique maximum C_{E} at $x = 0$ and $\mathbb{E}\{\theta_0 - U\} = 0$. Thus U is concentrated on the integer coset $\theta_0 + \mathbb{Z}$, so there exist phases ϑ_n such that

$$\mathbb{E}\,\mathrm{dist}(S - \vartheta_n, \mathbb{Z}) \longrightarrow 0. \tag{5.9}$$

To deduce $\mathcal{B}(X_n) \to 1$ from (5.9), let $q = q_*$, $t = 2q - 1 = t_*$, and let $P_q(u)$ be the 1-periodic sawtooth function equal to $-2(1-q)u$ on $[0, q]$ and $2q(u-1)$ on $[q, 1]$. Then $P_q(x - \vartheta)$ has mean $-q(1-q)$ on the circle and vanishes on $\vartheta + \mathbb{Z}$ (with $|P_q(x-\vartheta)| \le 2\,\mathrm{dist}(x-\vartheta, \mathbb{Z})$), while the combined test function $-tx + P_q(x - \vartheta)$ is 1-Lipschitz on $\mathbb{R}$. Using $\mathbb{E}S^* = \kappa/(2V)$ and $|\mathbb{E}e^{iuS^*}| \le 1/(|u|\sqrt{V})$ in the Fourier series of $P_q(x - \vartheta)$ gives, by Kantorovich–Rubinstein duality,

$$W_1(S, S^*) \ge \frac{t\kappa}{2V} + q(1-q) - 2\mathbb{E}\,\mathrm{dist}(S - \vartheta, \mathbb{Z}) - \frac{\zeta(3)}{\pi^3\sqrt{V}}. \tag{5.10}$$

Taking $\liminf$ via (5.9) gives $\liminf W_1(S, S^*) \ge t_*^2/2 + p_*q_* = r_*/2$, which matches the upper bound $W_1(S, S^*) \le P/(2V) \to r_*/2$ and proves $\mathcal{B}(X_n) \to 1$.

Finally, (5.9) implies $|\mathbb{E}e^{2\pi\mathrm{i}S}| = \prod_i |\mathbb{E}e^{2\pi\mathrm{i}Z_i}| \to 1$, so $Q = \sum_i \mathbb{E}Z_i^2 \le \frac{1}{4}\sum_i(1 - |\mathbb{E}e^{2\pi\mathrm{i}Z_i}|^2) \to 0$. Since $|Z_i| \le 1$, the direct couplings (Y_i, J_i) satisfy $\sum_i W_2^2(Y_i, J_i) \le Q \to 0$ and $\sum_i W_3^3(Y_i, J_i) \le Q \to 0$. Restoring the gap scale h_n proves (1.4) and completes the proof of Theorem 1.1.

Remark 5.4. While $\mathcal{B}(X_n) \to 1$ whenever $L_n \to 0$ and $C_n \to C_{\mathrm{E}}$, the pointwise inequality $C \le C_{\mathrm{E}}\mathcal{B}(X)$ does not hold for general distributions. Indeed, for $N = 1$, replacing the standard normal law Z on $[-a, a]$ by three atoms at $\{-a, 0, a\}$ with matching mean and second moment (weights $Pr/2, P(1-r), Pr/2$ where $P = \mathbb{P}(|Z| \le a)$ and $r = \mathbb{E}[Z^2\mathbf{1}_{\{|Z|\le a\}}]/(a^2P) \to 1/3$) gives $d_{\mathrm{K}} = P(1-r)/2$ and $W_1 = (aP/2)\{(1-r)^2 + r^2\}$, so $C/\mathcal{B}(X) = (1-r)/[2a\{(1-r)^2 + r^2\}] \sim 3/(5a) \to \infty$ as $a \downarrow 0$.

6 Variational contacts and uniform phase confinement

For the remainder of the paper, each array in the sequence is a global maximizer of C_{N_n} at its number of summands N_n and satisfies $L_n \to 0$. Since the number of nonzero summands is at least $L_n^{-2} \to \infty$, we have $N_n \to \infty$, and comparing with an N_n-summand i.i.d. Esseen array gives $C_{N_n} \to C_{\mathrm{E}}$, so Theorem 1.1 applies.

Lemma 6.1 (Variational contact and support coercivity)**.** *Normalize total variance to $B = 1$, so $R = L$. For all sufficiently late global maximizers, every support point of every summand satisfies $|u| \le 246R^{1/3}$. At an attaining positive closed threshold z, with $D = F(z) - \Phi(z)$, $\alpha = \phi(z)$, $A_0 = (3D + z\alpha)/2$, $v = \mathbb{E}X_i^2$, $\rho = \mathbb{E}|X_i|^3$, and $M_i = \mathbb{E}[X_i|X_i|]$, every summand X_i satisfies*

$$F_{-i}(z-u) + (\alpha + 3CM_i)u + A_0u^2 - C|u|^3 \le F(z) + A_0v - C\rho \qquad (u \in \mathbb{R}), \tag{6.1}$$

with equality at every $u \in \operatorname{supp} X_i$.

Proof. Perturbing $\mathrm{Law}(X_i)$ to $(1-e)\,\mathrm{Law}(X_i) + e\delta_u$ at the fixed closed threshold z and recentering and restandardizing gives (6.1) by maximality (adapting He and Cheng [7, Section 2.1, eqs. (7)–(12)] to a single coordinate), and upper semicontinuity of $F_{-i}(z-u)$ extends equality from a set of full $\mathrm{Law}(X_i)$-measure to $\operatorname{supp} X_i$. Substituting the deleted Berry–Esseen bound $|F_{-i}(z-u) - \Phi((z-u)/\sqrt{1-v})| \le (123/10)R$ [4, Theorem 2.1] into the equality at $u = R^{1/3}w \in \operatorname{supp} X_i$ gives $(99/500)w^3 \le 1023/40 + (129/10)w + (203/20)w^2$, which forces $w < 246$. □

In the gap units of Lemma 4.1, write $Y_i = J_i + Z_i$ with $V = \sum_i \mathbb{E}Y_i^2$, $d = V^{-1/2}$, $P = \sum_i \mathbb{E}|Y_i|^3$, $Q = \sum_i \mathbb{E}Z_i^2$, $T = \sum_i Z_i$, and standardized threshold $z = y/\sqrt{V}$. By Theorem 1.1 and Lemma 6.1, $|Y_i| \le H = O(V^{1/3})$, $\sup_i \mathbb{E}|Y_i|^3 = o(V)$, and

$$\frac{P}{V} \to r_*, \qquad \frac{\kappa}{V} \to t_*, \qquad Q \to 0, \qquad z \to 0, \qquad \frac{D}{d} \to \phi(0)\left(\frac{1}{2} + \frac{t_*}{6}\right), \qquad \mathrm{Cov}(K, T) = o(\sqrt{V}). \tag{6.2}$$

Lemma 6.2 (Uniform support-phase confinement)**.** *For the decomposition $Y_i = J_i + Z_i$,*

$$\eta_n := \max_{1 \le i \le N_n} \|Z_i\|_\infty \longrightarrow 0, \tag{6.3}$$

and the fractional part $\{y + \mu\}$ converges to 0 modulo 1. In particular, whenever $t := \mathbb{E}|T| > 0$, we have $Q = o(t)$.

Proof. For any deleted coordinate i, put $v_i = \mathbb{E}Y_i^2$, $\rho_i = \mathbb{E}|Y_i|^3$, $w_i = V - v_i$, and $\gamma_i = \kappa(Y_{-i})/w_i$, where $\kappa(Y_{-i})$ is the deleted third-moment sum. For $|u| \le H$, if the deleted fractional part $\tau_i(u) = \{y - u + \mu_{-i}\}$ lies in $[\epsilon, 1 - \epsilon]$, then Lemma 5.1 applied to the deleted sum gives

$$F_{-i}(y - u) = G_i(u) - d\phi(0)\tau_i(u) + o(d), \tag{6.4}$$

where $G_i(u) = \Phi(x_i) + w_i^{-1/2}\phi(x_i)\{1/2 + \gamma_i(1 - x_i^2)/6\}$ ($x_i = (y - u)/\sqrt{w_i}$) is the smooth deleted Edgeworth approximation. In gap units, the contact score and level from (6.1) are

$$g_i(u) = F_{-i}(y - u) + b_i u + A_0 u^2 - \lambda|u|^3, \qquad \ell_i = F(y) + A_0 v_i - \lambda\rho_i,$$

where

$$b_i = d\phi(z) + 3\lambda\mathbb{E}[Y_i|Y_i|], \qquad \lambda = \frac{D}{P} = Cd^3, \qquad A_0 = \frac{3D + z\phi(z)}{2V},$$

so that $g_i(u) \le \ell_i$ for all $u \in \mathbb{R}$ with equality on $\operatorname{supp} Y_i$. Expanding around $u = 0$ shows that uniformly in i and $|u| \le H$,

$$G_i(u) + b_i u + A_0 u^2 - \lambda|u|^3 - \ell_i \le o(d) \tag{6.5}$$

(since $C \to C_{\mathrm{E}} > \phi(0)/6$ makes the cubic coefficient $-\lambda + \phi(0)d^3/6$ strictly negative). If $u \in \operatorname{supp} Y_i$ had $u + \mathbb{E}K_i$ bounded away from $\mathbb{Z}$, then $\tau_i(u)$ would stay in $[\epsilon, 1 - \epsilon]$ (since $\{y + \mu\} \to 0$) and (6.4)–(6.5) would give $g_i(u) < \ell_i$, contradicting $u \in \operatorname{supp} Y_i$. Thus $\eta_n = \max_i \|Z_i\|_\infty \to 0$. Finally, $\mathbb{E}T^4 \le 3Q^2 + \eta_n^2 Q$ and $t = \mathbb{E}|T| \ge Q^{3/2}/(\mathbb{E}T^4)^{1/2} \ge Q/\sqrt{3Q + \eta_n^2}$ imply $Q = o(t)$ whenever $t > 0$. □

7 Relative conditional first-moment bounds

To compare the global maximizer against its exact integer rounding J_i when $t = \mathbb{E}|T| > 0$ is arbitrarily small, we first adjust the common scale by $1 + \delta = 1 + o(A^{-1/2})$ to place the full characteristic product at its exact local maximum near 2π, keeping the integer labels K_i unchanged.

Lemma 7.1 (Alignment at the exact full-product peak)**.** *After a common rescaling by $1 + o(A^{-1/2})$, the summands $Y_i = J_i + Z_i$ retain the same integer parts J_i and satisfy (6.3) and*

$$\begin{gathered} A = \operatorname{Var} K \to \infty, \qquad |J_i| \le H = O(A^{1/3}), \\ \sum_{i=1}^{N} \mathbb{E}|J_i|^3 = O(A), \qquad Q \to 0, \qquad |\operatorname{Cov}(K, T)| \le KHQ. \end{gathered} \tag{7.1}$$

Proof. Because $|\mathbb{E}e^{\mathrm{i}(2\pi+x/\sqrt{A})\sum_i J_i}| \to e^{-x^2/2}$ and $T \to 0$ in L^1, the full product $|\mathbb{E}e^{\mathrm{i}s\sum_i Y_i}|$ attains a local maximum at $s = 2\pi(1+\delta)$ with $\delta = o(A^{-1/2})$. Rescaling Y_i by $1+\delta$ and realigning the phases $\beta_i = (2\pi)^{-1}\arg\mathbb{E}e^{2\pi\mathrm{i}Z_i^{\mathrm{new}}}$ keeps the nearest-integer labels K_i unchanged, while stationarity $\sum_i \mathbb{E}[Y_i^{\mathrm{new}}\sin(2\pi E_i)]/r_i = 0$ at the full peak yields $|\operatorname{Cov}(\sum_i Y_i^{\mathrm{new}}, T_{\mathrm{new}})| \le KHQ_{\mathrm{new}}$, proving (7.1). □

Lemma 7.2 (Multiplier bound in terms of $t = \mathbb{E}|\sum_i Z_i|$)**.** *Let $Z_1,\dots,Z_N$ be independent centered random variables with $t = \mathbb{E}|\sum_{i=1}^N Z_i|$. For any random variables $W_1,\dots,W_N$ such that each pair (W_i, Z_i) is independent of $(W_j, Z_j)_{j\neq i}$ and $\|W_i\|_\infty \le M_W$,*

$$\left|\sum_{i=1}^N \mathbb{E}[W_i Z_i]\right| \le 8t\left(\sum_{i=1}^N \mathbb{E}W_i^2\right)^{1/2} + 5M_W t. \tag{7.2}$$

Proof. Let Z_i' be an independent copy of Z_i (independent of W_i) and set $D_i = Z_i - Z_i'$. By Khintchine's inequality, $\mathbb{E}(\sum_i D_i^2)^{1/2} \le 2\sqrt{3}\,t =: C_0 t$. Truncating at $b = 4C_0 t$, Markov's inequality gives $\mathbb{P}(\max_{j\neq i}|D_j| \le b) \ge 3/4$, so summing over the disjoint events where exactly one $|D_i| > b$ yields $\sum_i \mathbb{E}[|D_i|\mathbf{1}_{\{|D_i|>b\}}] \le (4/3)C_0 t < 5t$. For $U = \sum_i \mathbb{E}[D_i^2\mathbf{1}_{\{|D_i|\le b\}}]$, second/fourth-moment interpolation gives $U/\sqrt{U+b^2} \le C_0 t$, so $U < 60t^2$. Applying $\mathbb{E}[W_i Z_i] = \mathbb{E}[W_i D_i]$, Cauchy–Schwarz on $\{|D_i| \le b\}$, and $\|W_i\|_\infty \le M_W$ on $\{|D_i| > b\}$ proves (7.2). □

Lemma 7.3 (Relative conditional first-moment bounds)**.** *Under the hypotheses of Lemma 7.1, whenever $t = \mathbb{E}|T| > 0$, for each fixed $B_0 > 0$ and uniformly over $|k-\mu| \le B_0\sqrt{A}$,*

$$c\,t \le \mathbb{E}(|T| \mid K = k) \le Kt, \qquad |\mathbb{E}(T \mid K = k)| = o(t). \tag{7.3}$$

Proof. Take two independent copies (K,T) and (K',T') of the row, with $D_i = Z_i - Z_i'$ and $D = \sum_i D_i$. Fix a large constant $B > 1$, truncate at $b = Bt$, and set $d_i = D_i\mathbf{1}_{\{|D_i|\le b\}}$, $W = \sum_i d_i$, $U = \sum_i \mathbb{E}d_i^2$, and $\Delta = \sum_i \mathbb{E}[|D_i|\mathbf{1}_{\{|D_i|>b\}}]$. As in the proof of Lemma 7.2,

$$\Delta \le Kt, \qquad \sum_{i=1}^N \mathbb{P}(|D_i| > b) \le \frac{K}{B}, \qquad U \le K_B t^2 \quad (K_B = O(B)). \tag{7.4}$$

On the frequency cell $\|s\| \le c/H$ ($s = (s_1, s_2)$), set

$$f_i(u) = \mathbb{E}e^{\mathrm{i}uJ_i}, \qquad m_{ir}(s) = \frac{\mathbb{E}[d_i^r e^{\mathrm{i}(s_1 J_i + s_2 J_i')}]}{f_i(s_1)f_i(s_2)}.$$

Applying Lemma 7.2 to the antisymmetric variables d_i bounds the linear twisted sum by

$$\left|\sum_{i=1}^N m_{i1}(s)\right| \le Kt\Big\{H\|s\| + (\sqrt{HA}+H^2)\|s\|^2 + (H^2\sqrt{A}+H^3)\|s\|^3\Big\}, \tag{7.5}$$

which is $o(t)$ on the Gaussian scale $\|s\| = O(A^{-1/2})$. Fourier inversion of the second and fourth moments of W (using (5.5) outside $\|s\| \le c/H$) yields, uniformly for $|k_1-\mu|, |k_2-\mu| \le B_0\sqrt{A}$,

$$\mathbb{E}(W^2 \mid K = k_1, K' = k_2) = U + o(t^2), \qquad \mathbb{E}(W^4 \mid K = k_1, K' = k_2) \le C_B t^4. \tag{7.6}$$

For the discarded tail $D - W = \sum_i D_i\mathbf{1}_{\{|D_i|>b\}}$, one-pair deletion ratios $\mathbb{P}(K_{-i} = k - k_{i1}, K'_{-i} = k - k_{i2})/\mathbb{P}(K = K' = k) = 1 + o(1)$ uniformly over shifts $O(H)$ imply $\mathbb{E}(|D-W| \mid K = K' = k) \le 2\Delta \le 2Kt$, while (7.6) and $U \le K_B t^2$ give $\mathbb{E}(|W| \mid K = K' = k) \le K(B)t$, so $\mathbb{E}(|D| \mid K =$

$K' = k) \le K'(B)t$. For the lower bound, if $\Delta \ge \epsilon t$, conditioning on the disjoint events where a single $|D_i| > b$ gives $\mathbb{E}(|D| \mid K = K' = k) \ge c\Delta \ge c\epsilon t$ for large B; if $\Delta < \epsilon t$, then $U \ge (1-\epsilon)^2 t^2$ and moment interpolation in (7.6) gives $\mathbb{E}(|W| \mid K = K' = k) \ge c_B t$, so choosing $\epsilon < c_B/4$ gives $2\mathbb{E}(|T| \mid K = k) \ge \mathbb{E}(|D| \mid K = K' = k) \ge c_B t/2$. Finally, applying the one-copy Fourier expansion with (7.5) and $|\operatorname{Cov}(K,T)| \le KHQ = o(Ht)$ gives $|\mathbb{E}(T \mid K = k)| = o(t)$, and conditional Jensen yields

$$\mathbb{E}(|T| \mid K = k) \le \mathbb{E}(|T - T'| \mid K = K' = k) + |\mathbb{E}(T \mid K = k)| \le K''(B)t.$$

□

8 Exact phase collapse for global maximizers

For the central integer layer $k = \operatorname{round}(y + \mu)$ and offset $\theta = y + \mu - k \to 0$, define

$$w(\theta) = \mathbb{P}(T > \theta \mid K = k), \qquad f(\theta) = \theta + w(\theta).$$

Lemma 8.1 (Relative closed-threshold loss)**.** *If $t = \mathbb{E}|T| > 0$, then at the attaining central layer k and offset θ,*

$$f(\theta) \ge ct, \qquad |\theta| + w(\theta) + t \le Kf(\theta). \tag{8.1}$$

Proof. Truncating T to $X = T\mathbf{1}_{\{|T| \le 1/4\}}$ changes its conditional CDF, mean, and L^1 norm by $O(Q) = o(t)$ (via (5.4)), so $m = \mathbb{E}(X \mid K = k) = o(t)$ and $s = \mathbb{E}(|X| \mid K = k) \ge ct$ by Lemma 7.3. Put $u = \mathbb{P}(X > \theta \mid K = k)$. For $0 \le \theta \le 1/4$, $s = 2\mathbb{E}(X_+ \mid K = k) - m \le 2\theta + \frac{1}{2}u - m \le 2(\theta + u) - m$; for $\theta = -b \in [-1/4, 0)$, $m \le (1/4 + b)u - b \le u/2 - b$ and $s \le u/2 - m$ give $\theta + u \ge s + 2m$, proving (8.1). □

Proposition 8.2 (Exact collapse of rounding residuals)**.** *For every sufficiently late global maximizer in the sequence, $T = 0$ almost surely, so every summand $Y_i = J_i$ is supported on a single coset of $\mathbb{Z}$.*

Proof. Suppose $t = \mathbb{E}|T| > 0$ along a subsequence. Compare the global maximizer $(Y_1, \dots, Y_N)$ directly against the centered integer array $(J_1, \dots, J_N)$ at the same number of summands N and integer threshold $k - \mu$. By (5.4), decomposing $\{S \le k + \theta - \mu\}$ by the value of K gives

$$F_Y(k + \theta - \mu) = F_K(k) - p_k w(\theta) + E_{\text{cross}}, \qquad |E_{\text{cross}}| \le KdQ = o\big(df(\theta)\big). \tag{8.2}$$

Since $p_k = d\phi(z) + o(d)$, $\Phi((k + \theta - \mu)/\sqrt{V}) = \Phi((k - \mu)/\sqrt{A}) + d\phi(z)\theta + o(df(\theta))$ (using $|V - A| \le KHQ = o(At)$), and Lemma 8.1 controls both $|\theta|$ and $w(\theta)$ by $f(\theta)$, the discrepancies satisfy

$$D_Y = D_J - d\phi(z)f(\theta) + o\big(df(\theta)\big). \tag{8.3}$$

Along the interpolation $Y_i(s) = J_i + sZ_i$ $(0 \le s \le 1)$, Lemma 7.2 with $W_i = 3Y_i(s)|Y_i(s)|$ bounds $|\frac{d}{ds}\sum_i \mathbb{E}|Y_i(s)|^3| \le K(\sqrt{HA} + H^2)t = o(At)$, so $|P_Y - P_J| = o(At)$. Since $V \sim A$ $(d = V^{-1/2})$ and $P_Y \sim P_J \sim r_* V$, the prefactors $V^{3/2}/P_Y \sim (r_* d)^{-1}$ and $A^{3/2}/P_J$ differ by a relative error of $o(t)$, that is, $|V^{3/2}/P_Y - A^{3/2}/P_J| = o(d^{-1}t)$. Because $D_J = O(d)$ and $t \le Kf(\theta)$, multiplying (8.3) by $V^{3/2}/P_Y$ shows that $(J_1, \dots, J_N)$ has Berry–Esseen ratio strictly larger than that of $(Y_1, \dots, Y_N)$ by at least $c' f(\theta) \ge c'' t > 0$, contradicting global maximality of $(Y_1, \dots, Y_N)$. Thus $t = 0$ eventually, so $Z_i = 0$ almost surely for all i. □

9 Discrete concavity and reduction to two adjacent lattice points

By Proposition 8.2, for all sufficiently late global maximizers, each summand Y_i is supported on a single coset of $\mathbb{Z}$.

Lemma 9.1 (Reduction to two adjacent lattice points)**.** *For all sufficiently late global maximizers, every nonconstant summand Y_i is supported on two adjacent points of its lattice coset.*

Proof. Let $c_i(x) = \mathbb{P}(\sum_{j\neq i} Y_j = x)$ be the deleted point mass function on its coset, with deleted variance $w_i = V - v_i$, $d_i = w_i^{-1/2}$, deleted signed cubic sum $\kappa_i = \kappa(Y_{-i})$, and $H_3(x) = x^3 - 3x$. Fourier inversion with multiplier $e^{-\mathrm{i}u} - 1$ and (5.5) gives, uniformly in i and x,

$$c_i(x+1) - c_i(x) = d_i^2\phi'(d_i x) + \frac{d_i^3}{2}\phi''(d_i x) + d_i^3\frac{\kappa_i}{6w_i}(H_3\phi)'(d_i x) + o(d^3). \tag{9.1}$$

For the contact function $g_i(u) = F_{-i}(y-u) + b_i u + A_0 u^2 - \lambda|u|^3$ on the coset of Y_i, its centered second difference $\Delta^2 g_i(u) = g_i(u+1) - 2g_i(u) + g_i(u-1)$ has deleted-CDF contribution $c_i(y-u+1) - c_i(y-u)$, whose leading $-d^2 z\phi(z)$ term cancels the $d^2 z\phi(z)$ part of $2A_0$, leaving for $|u| \le H+1$:

$$d^{-3}\Delta^2 g_i(u) = \phi(0)(1+u) - C_{\mathrm{E}}\{|u+1|^3 + |u-1|^3 - 2|u|^3\} + o(1+|u|) + O(d|z|v_i). \tag{9.2}$$

Because $C_{\mathrm{E}} > \phi(0)$ and $|u+1|^3 + |u-1|^3 - 2|u|^3$ equals $6|u|$ for $|u| \ge 1$ and $2 + 6u^2 - 2|u|^3 \ge 2 + 4u^2$ for $|u| \le 1$, the leading expression in (9.2) is bounded above by $-c(1+|u|)$ with $c > 0$. Applying the discrete Green identity on any interval of length $L \ge 2$ between two contact points $u_0, u_0 + L \in \operatorname{supp} Y_i$,

$$g_i(u_0 + j) - g_i(u_0) = -\sum_{k=1}^{L-1} \frac{\min(j,k)(L - \max(j,k))}{L}\Delta^2 g_i(u_0 + k) \le 0,$$

first bounds L and v_i by a uniform constant (making $d|z|v_i = o(1)$ in (9.2)), and then implies $\Delta^2 g_i(u) < 0$ strictly at every coset point, which makes $L \ge 2$ impossible. Thus $L = 1$ for every nonconstant summand. □

Since every nonconstant summand is supported on two points at distance 1, all active summands share a common support diameter. Applying Theorem 1.1 of Bell [1] at the same horizon N_n shows that all active summands are identically distributed, completing the proof of Theorem 1.2.

10 Application to arrays containing a large two-point group

As an immediate illustration of Theorem 1.2, suppose a global maximizer of C_{M+r_M} contains $M \to \infty$ active two-point summands sharing a support diameter $h_M > 0$ with probabilities in a fixed compact set $J \subset (0,1)$, and that the remaining r_M summands satisfy

$$R_{\mathrm{rem}} := \sum_{i \notin \text{group}} \mathbb{E}|X_i|^3 = o(M^{3/2}h_M^3). \tag{10.1}$$

Because the two-point group contributes variance at least $c_J M h_M^2$ and third absolute moment sum at most KMh_M^3, the full array has Lyapunov ratio

$$L \le \frac{KMh_M^3 + R_{\mathrm{rem}}}{(c_J M h_M^2)^{3/2}} \longrightarrow 0.$$

Theorem 1.2 therefore implies that for all sufficiently large M, the entire $(M + r_M)$-summand global maximizer is a padded i.i.d. two-point array of common diameter h_M. In particular, if all r_M remaining summands have support diameter at most $H_M h_M$ with $r_M H_M^3 = o(M^{3/2})$, then (10.1) holds automatically.

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